AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 26 · WORKBOOK

Image and Voice Pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected pipeline-evidence record.

Explain stages, trace a conflict, audit a qualification and design a revealing follow-up.

Keep one habit: Keep source references, extracted fields, lookup rules, final claims and unavailable evidence separate.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Text as Pieces

Recommendation Scores

Repair Bias in Data

W1 · Explain the pipeline mechanism

> Name source, transformation, task rule and output.

Your evidence record

- Worked conflict: transcript15 boxes versus inventory50, no original audio and no stated authoritative record; preserve the conflict rather than select with unsupported certainty. Separate shelf case: known phrase14, transcript40, replay available and inventory14 declared authoritative for lookup. The supplied image-reading example also has reference14 versus simulated OCR40. These are different evidence records, not one combined measurement.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the unresolved quantity conflict

> Do not invent replay or unstated source authority.

Your evidence record

- Worked conflict: transcript15 boxes versus inventory50, no original audio and no stated authoritative record; preserve the conflict rather than select with unsupported certainty. Separate shelf case: known phrase14, transcript40, replay available and inventory14 declared authoritative for lookup. The supplied image-reading example also has reference14 versus simulated OCR40. These are different evidence records, not one combined measurement.

- Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

  Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Restore the missing qualification

> Separate a bounded supported answer from broader unverified claims.

Your evidence record

- Worked conflict: transcript15 boxes versus inventory50, no original audio and no stated authoritative record; preserve the conflict rather than select with unsupported certainty. Separate shelf case: known phrase14, transcript40, replay available and inventory14 declared authoritative for lookup. The supplied image-reading example also has reference14 versus simulated OCR40. These are different evidence records, not one combined measurement.

- A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing stage-specific check

> State expected evidence, fixed conditions and what remains unavailable.

Your evidence record

- Worked conflict: transcript15 boxes versus inventory50, no original audio and no stated authoritative record; preserve the conflict rather than select with unsupported certainty. Separate shelf case: known phrase14, transcript40, replay available and inventory14 declared authoritative for lookup. The supplied image-reading example also has reference14 versus simulated OCR40. These are different evidence records, not one combined measurement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Apply a new case independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

If voice or image **input** is added, show each transformation and a safe fallback.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Speech **pipeline**s and multimodal **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Speech **pipeline**s and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A voice book helper transcribes speech, extracts preference, then ranks books.

Where can a wrong suggestion begin?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 26 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

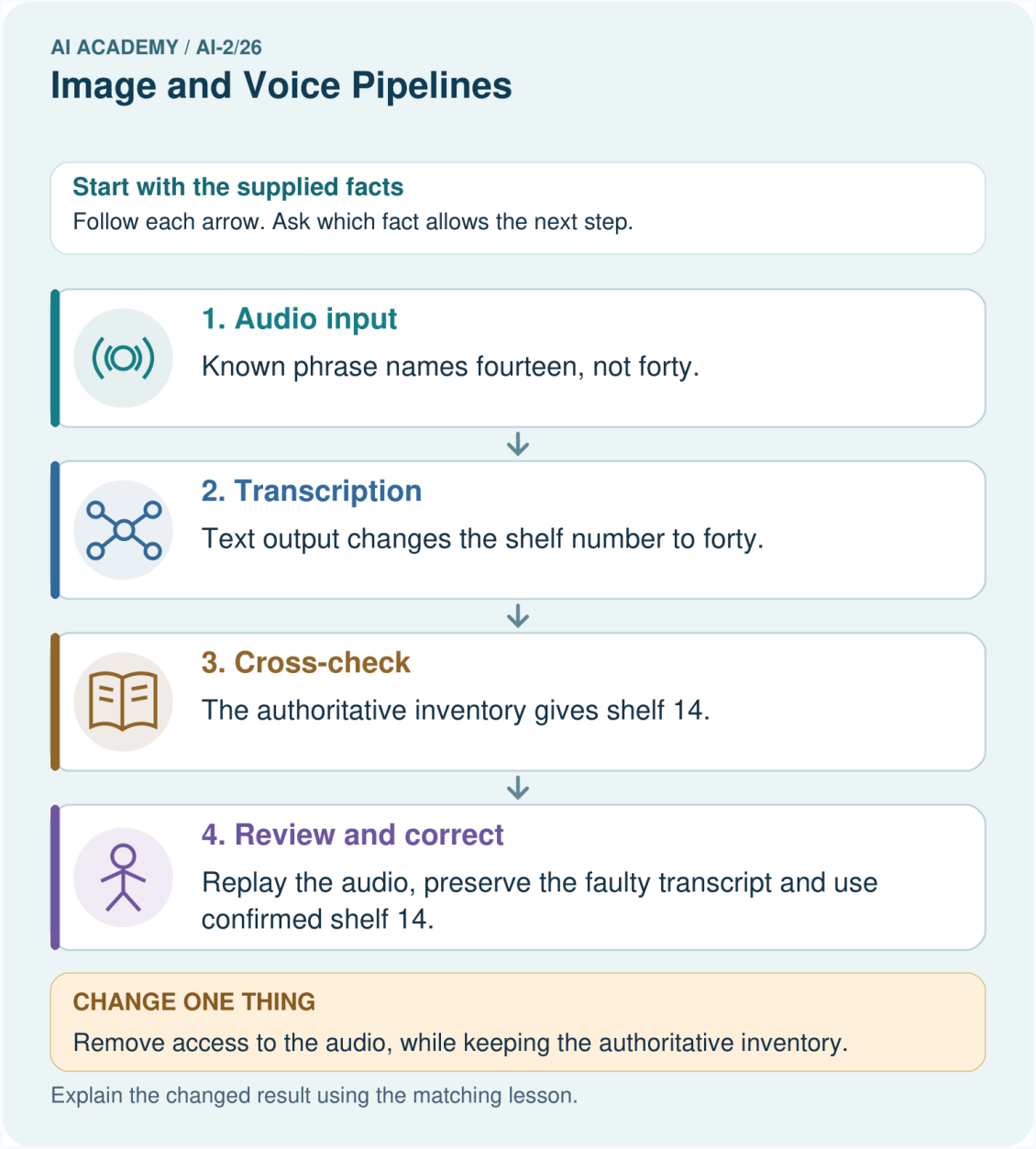
> Keep the explanation and the support you used.

Use this when helpful

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which stage first fails, and why should the faulty transcript not be silently overwritten?

> **My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.
- Which labelled step matches this detail? Replay the audio, preserve the faulty transcript and use confirmed shelf 14.
- 1. Audio **input**

- 2. Review and correct
- 3. **Transcription**
- 4. Cross-check
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Review and correct Replay the audio, preserve the faulty transcript and use confirmed shelf 14. A correct later lookup cannot repair an earlier wrong **input** by magic.

Locate the first difference between known audio and transcript, then compare the stated source authority. Keeping both records makes the correction auditable.

A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Changed-case question

- Changed-case practice: Remove access to the audio, while keeping the authoritative inventory. Which conclusion is supported by the supplied facts?
- 1. Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.
- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

The inventory resolves the shelf task; it does not recreate unavailable audio.

Repair and save your work

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

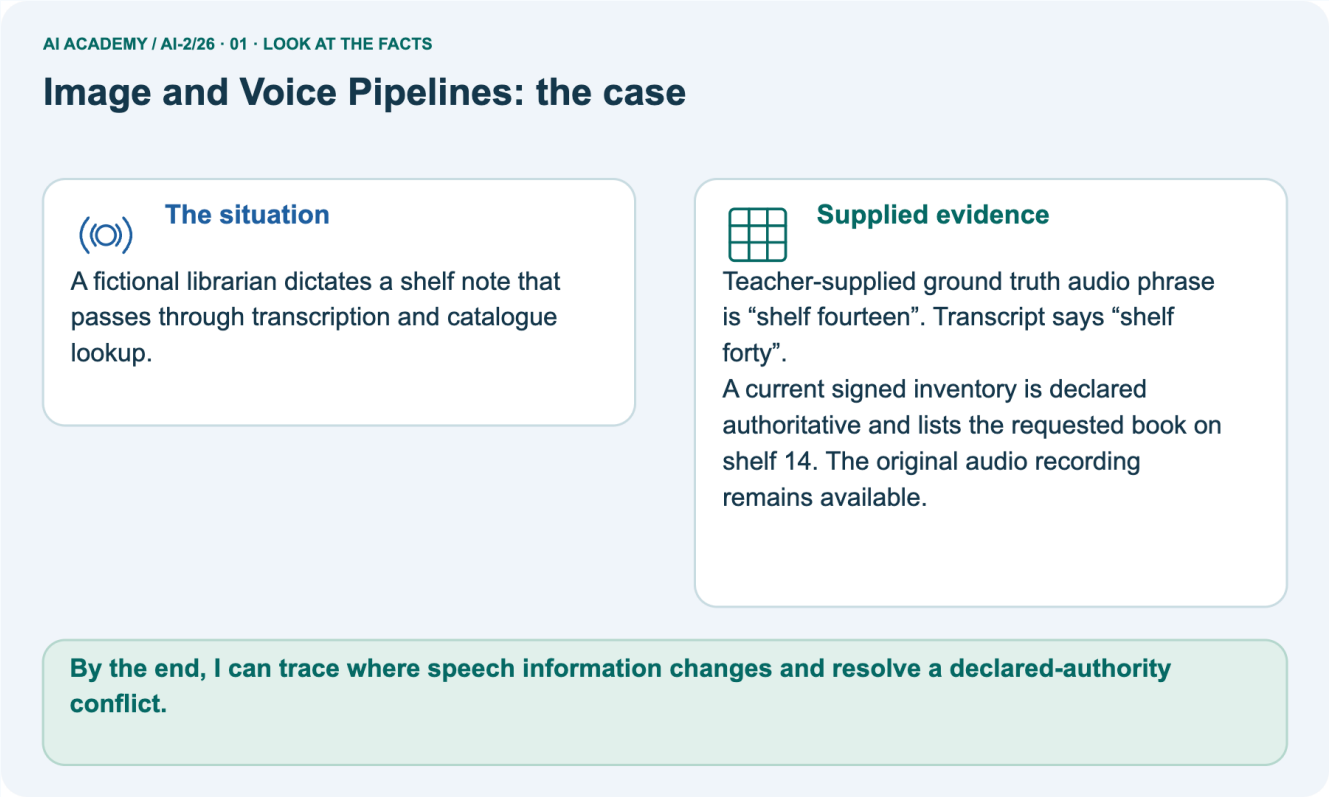
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Image and Voice Pipelines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

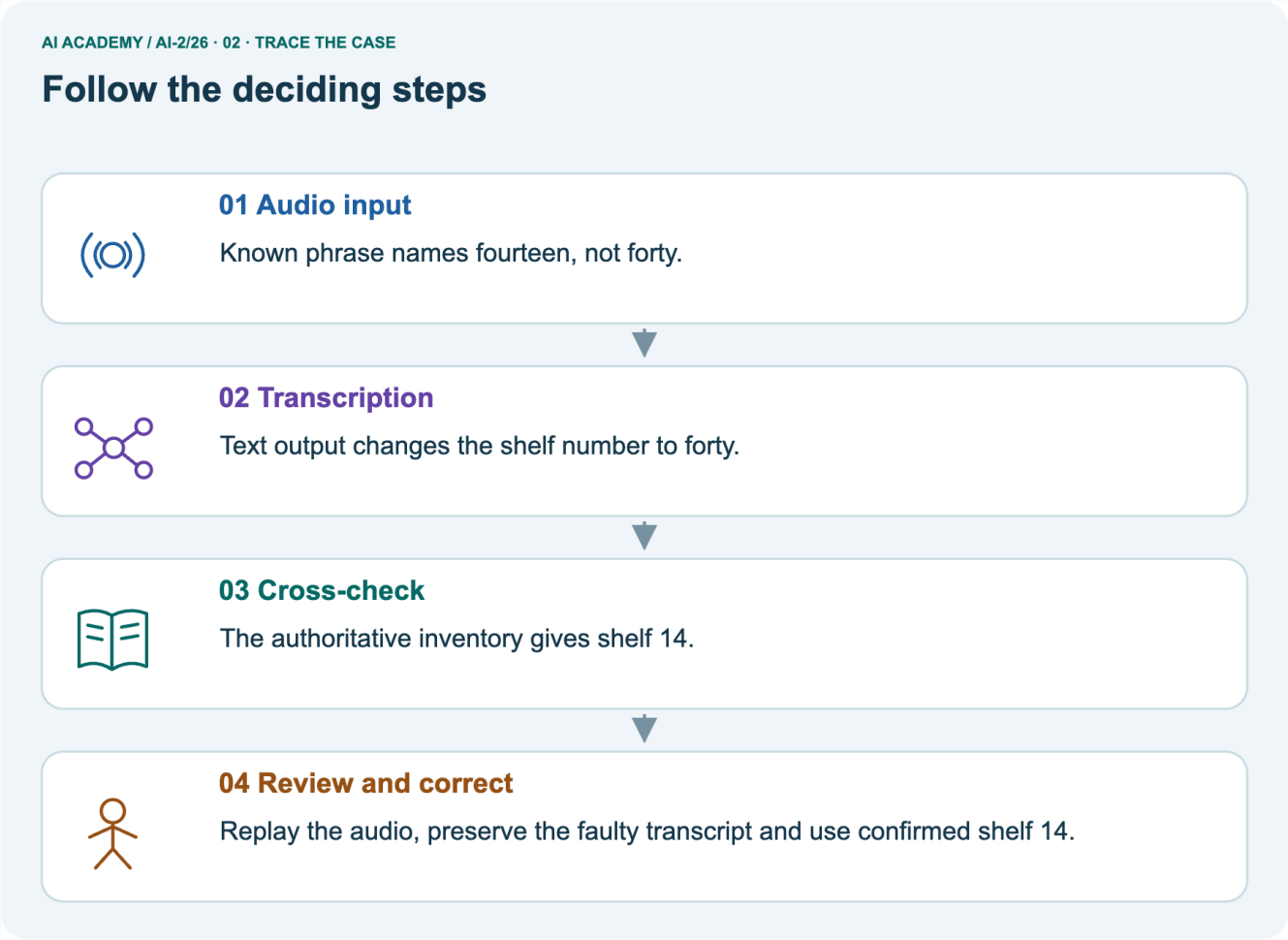
A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup. Supplied facts: Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



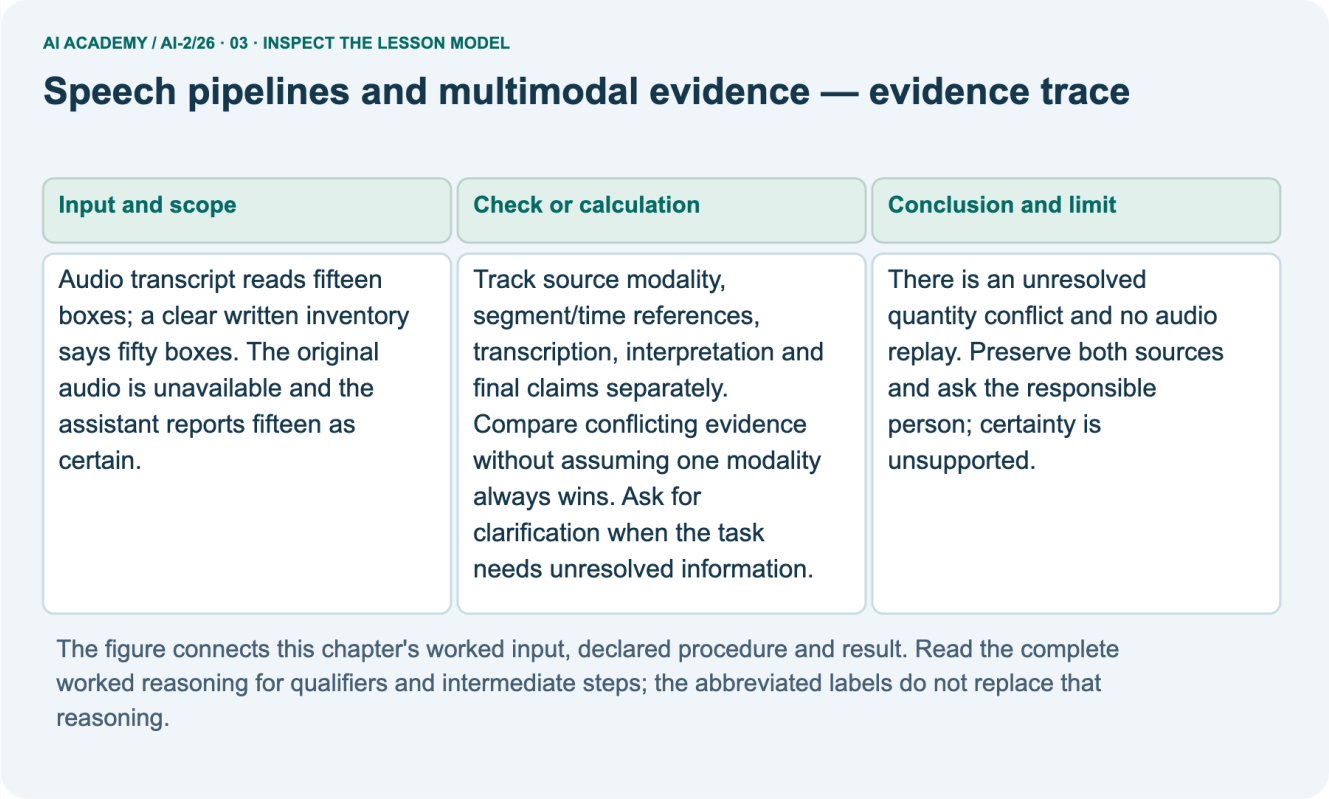
Follow the labelled picture with the matching explanation. Open the vector diagram

Audio **input**: Known phrase names fourteen, not forty. → **Transcription**: Text **output** changes the shelf number to forty. → Cross-check: The authoritative inventory gives shelf 14. → Review and correct: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech **pipeline**s and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

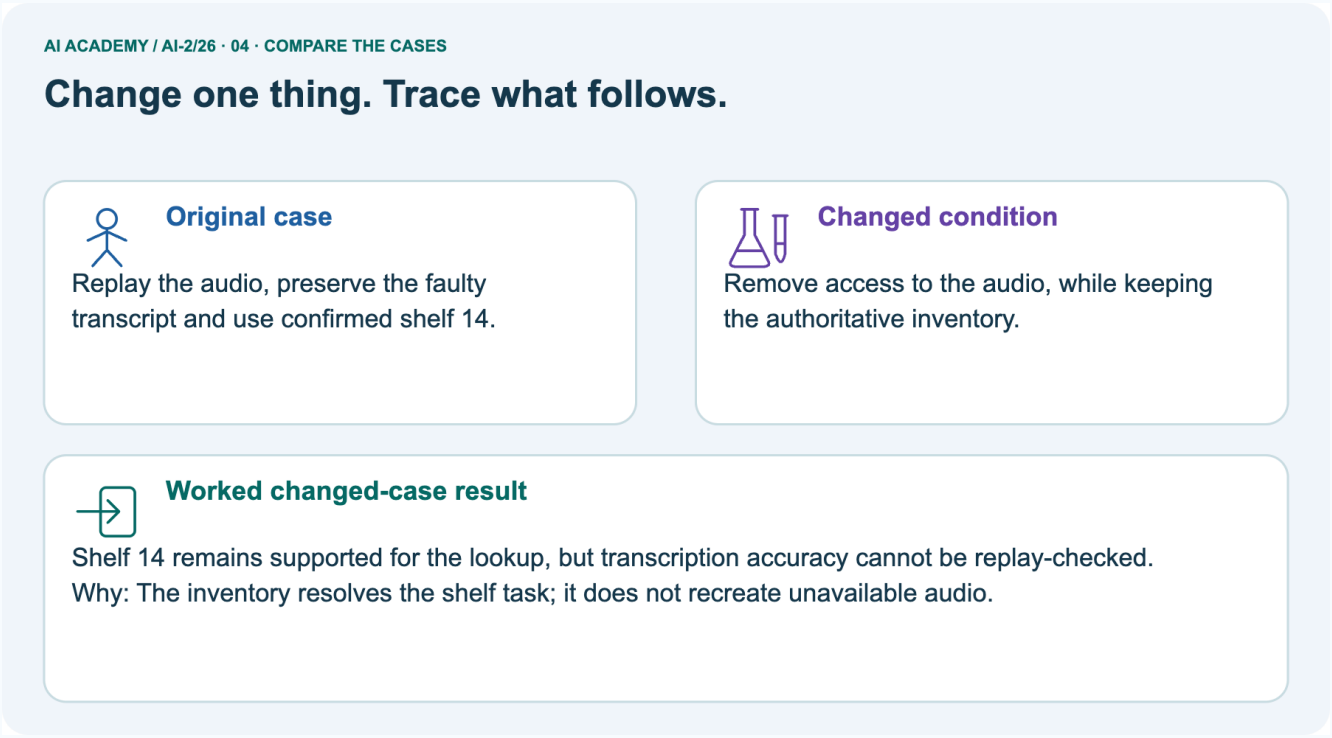
Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

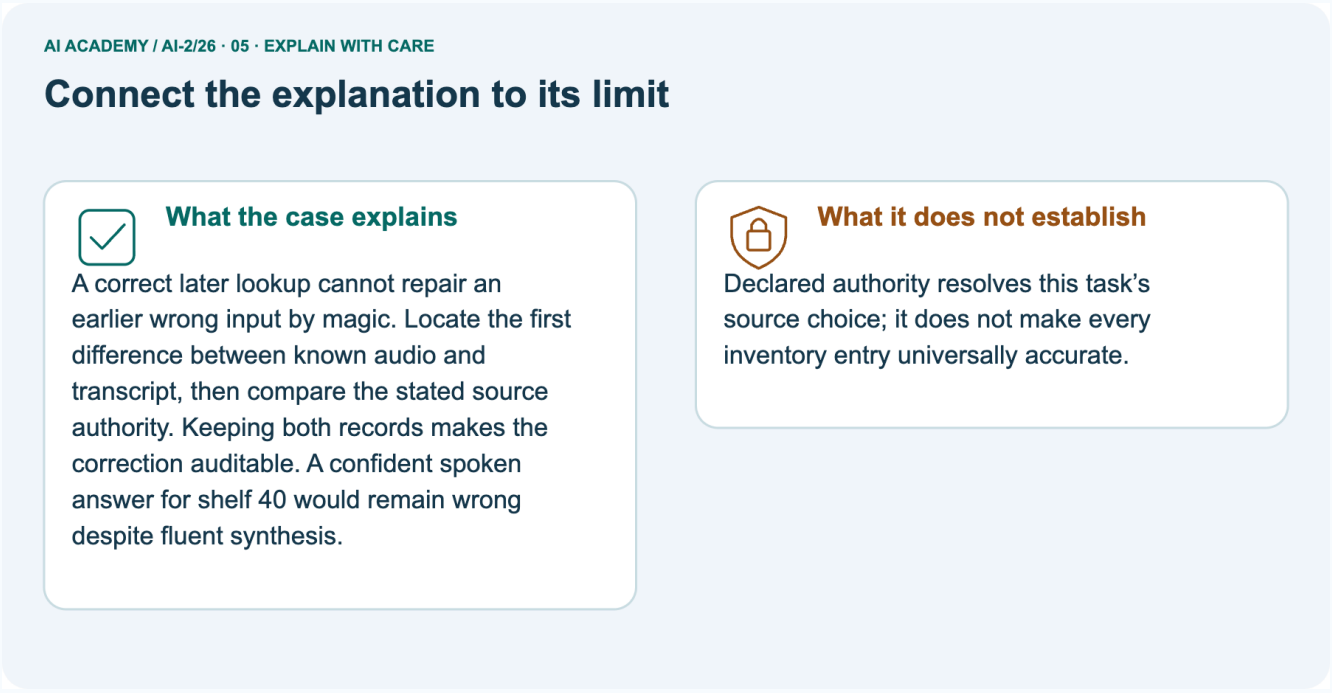
Original case: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.
Changed condition: Remove access to the audio, while keeping the authoritative inventory.

Expected result: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

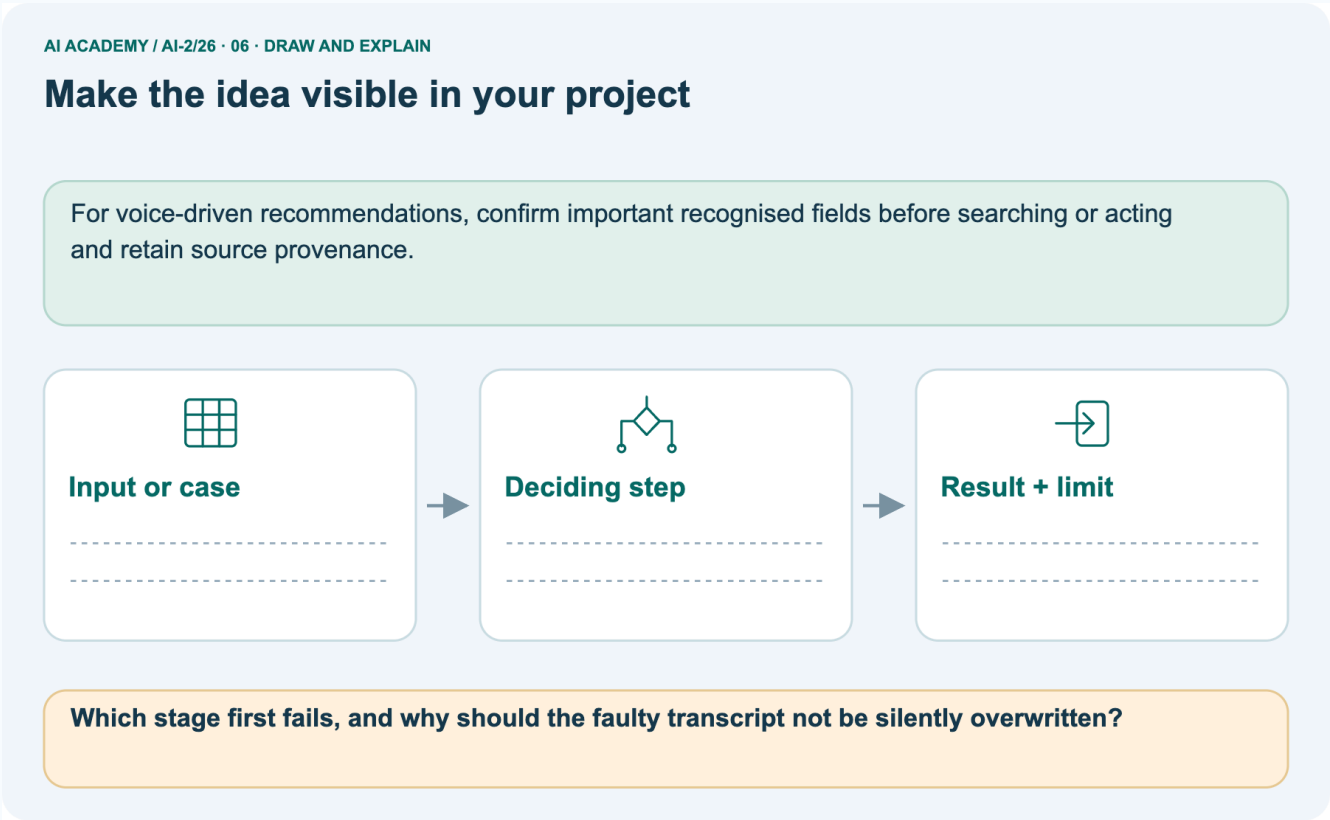
Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Limit: Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which stage first fails, and why should the faulty transcript not be silently overwritten?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Image and Voice Pipelines: the case

02. Follow the deciding steps

03. Speech pipelines and multimodal evidence — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.